

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 26-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-27** **Max. Marks: 180**

26-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2024-P2)_GTA-27_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

Rankers Academy JEE

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026


For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Let triangle ABC be an isosceles triangle with AB = AC. Suppose that the angle bisector of its angle B meets the side AC at a point D such that BC = BD + AD. Measure of the angle A in degrees, is

A) 80 B) 100 C) 110 D) 130

2. Find number of integers x, such that $x+\left[\frac{x}{6}\right]=\left[\frac{x}{2}\right]+\left[\frac{2x}{3}\right]$ which belongs to interval [31, 103]. ([.] represents greatest integer function)

A) 53 B) 56 C) 59 D) 65

3. If value of $S=\lim\limits_{n\to\infty}\frac{1}{n^4}\prod\limits_{i=1}^{2n}\left(n^2+i^2\right)^{1/n}$ is $\lambda\cdot e^{\mu\cdot\tan^{-1}\mu-4}$. Find $\lambda+\mu$.

A) 22 B) 24 C) 26 D) 27

4. If variance of an A.P. of first term 27, common difference 3, number of terms 15 is K times the smallest positive integer with exactly 24 positive integer divisors, then $\left[\frac{1}{K}\right]$ is ([.] represents greatest integer function)

A) 3 B) 0 C) 1 D) 2

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026


For More Join: @IITJEE_Advanced

5. Let $f:[1,\infty)\to R$ is a differentiable function such that $f'(x)=\dfrac{1}{x^2+f^2(x)}$ and $f(1)=1$, then which of the following is/are **CORRECT**?

A) $f(x)$ is strictly increasing in $(1,\infty)$

B) $\lim\limits_{x\to\infty} f(x)<2$

C) The number of solution of $f(x)=1+2x$ is 0

D) $f(3)<2$

6. Which of the following statements(s) is/are **CORRECT?**

A) An event A is known to be independent of the events B, $B\cup C$ and $B\cap C$, then A is also independent of C.

B) Let A^c denote the complement of an event A. If A and B are two events, then the probability that atmost one of A, B occurs is $P\left(A^c\right)+P\left(B^c\right)-P\left(A^c\cap B^c\right)$.

C) After a typist has written ten letters and had addressed the ten corresponding envelopes, a careless mailing clerk inserted the letters in the envelopes at random, one letter per envelop. The probability that exactly nine letters were inserted in the proper envelopes is $\dfrac{1}{10!}$

Rankers Academy JEE

D) An aircraft is equipped with three engines that operate independently. The probability of an engine failure is 0.01. If only one engine is needed for the successful operation of the aircraft, the probability of a successful flight, is 0.999999.7. Consider a parabola $S:x^2=-4(y-4)$ and a line $L:y=x+8$. 'S' intersects x–axis at A and B and L intersects x–axis at C such that C, A and B are collinear in order.

If a point 'P' is chosen on L such that angle APB is maximum and P is not collinear with A and B, then which of the following is/are **TRUE**? (B lies on positive x–axis)

A) PC = $\sqrt{48}$ B) PC = $\sqrt{32}$ C) PB = $\sqrt{3}$ PA D) PB = $\sqrt{2}$ PA

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

designated to enter the answer.
• Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

8. Let $\omega \neq 1$, be a 13th root of unity, find the remainder when $\prod_{k=0}^{12}\left(2-2\omega^{k}+\omega^{2k}\right)$ is divided by 1000.

9. If $\int_{0}^{1}\left(\frac{1-x^2}{1+x^2}\right)^2 \cdot \frac{\ell og\left(1+x^2\right)}{1+x^2}dx = \frac{a\cdot\pi\cdot\ell n2+\pi+b.I+c}{d}$ where $I=\int_{0}^{1}\frac{\ell ogx}{1+x^2}\cdot dx$ and a,b,c,d are integers, then find $a+b+c+d$.

10. For system of equations in (x,y) given by: $\left(x^2+xy+y^2\right)\cdot\sqrt{x^2+y^2}=88$, $\left(x^2-xy+y^2\right)\cdot\sqrt{x^2+y^2}=40$, if number of ordered pairs (x,y) satisfying is α also if β is the sum of all absolute values of x, then $[\alpha+\beta]$ is: (where [.] represents greatest integer function)

11. The value of K, for which $\int_{0}^{8/3}\left|x\cdot(8-3x)-K\right|\cdot dx$ is minimum, equals

12. Let A and C be two points on the line $\frac{x-1}{3}=\frac{y-2}{2}=\frac{z-3}{1}$ and B and D be two point on the line $\frac{x-2}{5}=\frac{y-1}{2}=\frac{z-2}{3}$ such that AB is perpendicular to both the given lines and the length of CD is $3\sqrt{3}$. If V is the maximum volume of the tetrahedron ABCD, then [V] is equal to (where [.] represents greatest integer function)

13. If $\sum_{r=0}^{10}\frac{{}^{10}C_r}{{}^{30}C_{10+r}}=\frac{{}^{n}C_m}{{}^{30}C_P\cdot{}^{P}C_m}$ then the value of $n+m+P$.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300/300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341/360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Let $\vec{a},\vec{b},\vec{c},\vec{d}$ are position vectors of 4 points A, B, C, D such that $|\vec{a}|=1,|\vec{b}|=2,|\vec{c}|=3,|\vec{d}|=4$ and $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d})=24,\ \vec{a}\cdot\vec{c}=\frac{3}{2},\ \vec{b}\cdot\vec{c}>0$, then area of $\Delta ABC=\frac{1}{4}\left(2\alpha+\beta\sqrt{3}\right)$ and area of $\Delta BCD=\frac{1}{2}\left(m-n\sqrt{3}\right)$, where $\alpha,\beta,m,n\in N$.

14. $3\alpha-\beta$ is equal to _____

15. $2m-n$ is equal to _____

PARAGRAPH-II

Let $A=\begin{bmatrix}\alpha & a & a\\ b & \beta & a\\ b & b & \gamma\end{bmatrix}$, $f(x)=\begin{vmatrix}\alpha+x & a+x & a+x\\ b+x & \beta+x & a+x\\ b+x & b+x & \gamma+x\end{vmatrix}$,

$g(x)=\alpha\beta\gamma-(\alpha\beta+\beta\gamma+\gamma\alpha)x+(\alpha+\beta+\gamma)x^2-x^3$,

$adj(A)=\left[a_{ij}\right]_{3\times3}, \sum_{i=1}^{3}\sum_{j=1}^{3}a_{ij}=18, |adj(A)|=576$

Based on above information, answer the following questions.

16. If $|A|>0$, then $f(30)+f(50)$ equals

17. If $|A|>0$, then $\frac{bg(a)-ag(b)+g(a)-g(b)}{(b-a)}$ equals

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300/300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341/360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

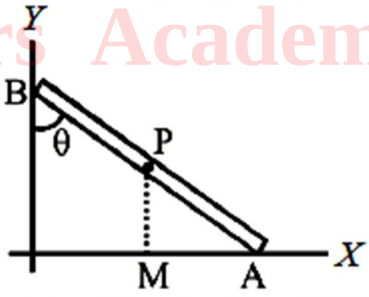
This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. An inductance L, a capacitance C and a resistance R may be connected to an AC source of angular frequency ω, in three different combinations of RC or RL or RLC in series. Assume that $\omega L = \frac{1}{\omega C}$. The power drawn by the three combinations are P_1, P_2, P_3 respectively. Then

A) $P_1 > P_2 > P_3$ B) $P_1 = P_2 < P_3$ C) $P_1 = P_2 > P_3$ D) $P_1 = P_2 = P_3$

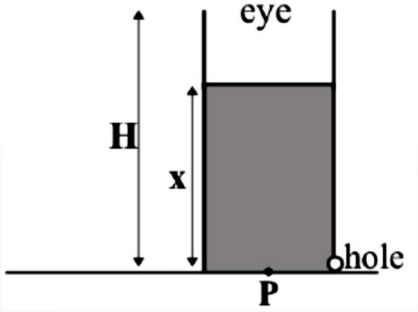
19. A rod of length ℓ is in motion such that its ends A and B are moving along x–axis and y–axis respectively. It is given that $\frac{d\theta}{dt} = 2 rad/s$ always. P is a fixed point on the rod. Let M be the projection of P on x–axis. For the time interval in which θ changes from 0 to $\frac{\pi}{2}$, choose the correct statement:-

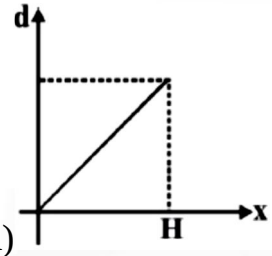


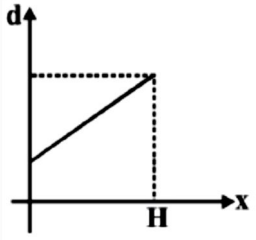
A) The acceleration of M is always directed towards right.
B) M executes SHM.
C) M moves with constant speed.
D) M moves with constant acceleration.

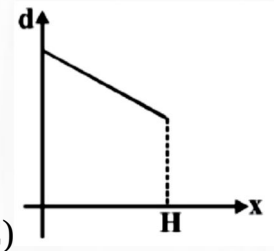
20. Liquid is filled in a vessel of height H. At the bottom of the vessel, there is a spot P and a hole to the wall from which liquid is coming out. Let d be the distance of image of P from an eye at height H from bottom at an instant when level of liquid in the vessel is x. If we plot a graph between d and x, it will be like

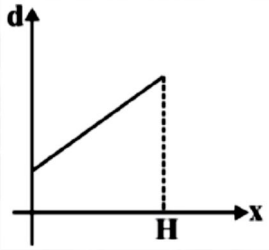
Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced



A) 

B) 

C) 

D) 

21. A projectile is projected from a horizontal ground at an angle of projection of 40°. After some time, it lands on the horizontal ground. During its motion:-

A) its distance from point of projection always increases.

B) its distance from point of projection first increases and then decreases.

C) velocity vector of projectile at some instant makes an angle 38° with its direction of acceleration.

D) velocity vector of projectile at some instant makes an angle 92° with its initial direction of motion.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

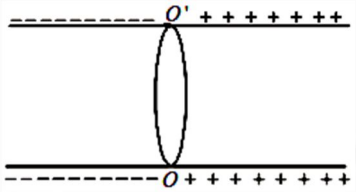
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class 300 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class 341 360
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class 720
For More Join: @IITJEE_Advanced

choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. Charges are distributed on the surface of an infinitely long hollow cylinder that the right half of the cylinder surface beginning from section OO' is charged positively, and the left half negatively. The value of surface charge density of the charges increases in the both directions and directly proportional to the distance from section OO'. Select the correct alternative(s).



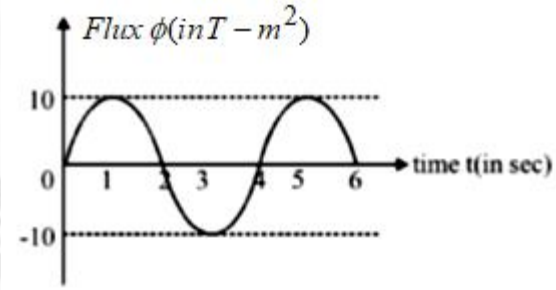
A) Electric field intensity is directed along the axis of the cylinder at all points inside the cylinder.

B) Electric field intensity is directed along the axis of the cylinder only on the points lying on the axis of the cylinder.

C) The magnitude of electric field intensity is same at all the points inside the cylinder.

D) The magnitude of electric field intensity is same only on the points lying on the axis of the cylinder.

23. A closed conducting loop, having resistance R, is being rotated about an axis perpendicular to the magnetic field. Magnetic flux through the closed conducting loop is continuously changing according to the graph shown in the figure. Then, which of the following statement(s) is/are correct?



A) The electric current through the loop is minimum (zero) at t = 1s, 3s and 5s.

B) The electric current through the loop is minimum (zero) at t = 0s, 2s and 6s.

C) Total charge flown through any cross–section of a closed conducting loop between 0 and 6s is zero.

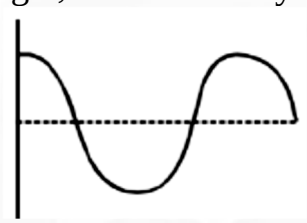
D) Total work done in rotating the loop in the magnetic field is zero in any time interval.

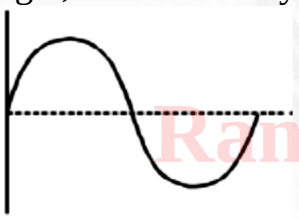
Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

24. There is a vertical spring on a railroad car at a station. The spring obeys Hooke's Law with spring constant 6000 N/m. You stand at rest on a light platform attached to the spring and it depresses by a distance 0.1 meter in equilibrium. Now if someone else displaces you vertically and then release at t = 0 such that your vertical displacement is represented by $y = A\cos(\omega t)$ meters, where $\omega = 10s^{-1}$, mark the **CORRECT** statement(s): (Assume g $= 10 m/s^2$)

A) Your mass is 60 kg.

B) The largest value of A for which you do not lose contact with platform is 0.1 m.

C) Now suppose you are oscillating with amplitude $A = 0.05m$ and the railroad car is moving to the right with constant velocity. Treating your equilibrium position at station as origin, as seen by someone standing still at the railroad station, your path is 

D) Now suppose you are oscillating with amplitude A = 0.05 m and the railroad car is moving to the right with constant velocity. Treating your equilibrium position at station as origin, as seen by someone standing still at the railroad station, your path is 

SECTION-III
(NON-NEGATIVE INTEGER)

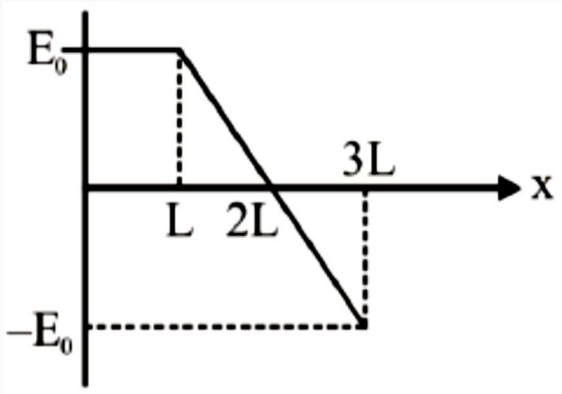
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases..*

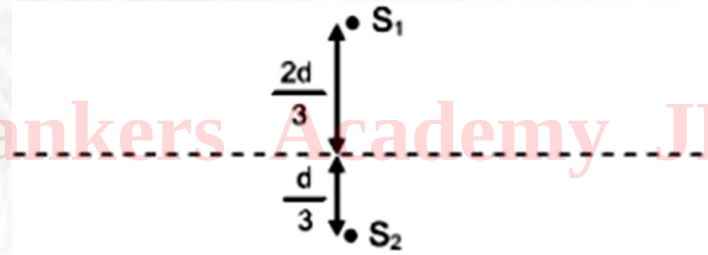
25. When the voltage applied to an X–ray tube is increased from $V_1 = 15.5\,kV$ to $V_2 = 31kV$, the wavelength interval between the K_α line and the short wavelength cut–off of the continuous X–ray spectrum increases by a factor of 1.3. Find the atomic number of the element of the target. If your answer is N, fill the value of $\frac{N}{13}$. (Take: $hc = 1240\ eV.nm$ and $R = 1\times10^7/m$)

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA 300/300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY 341/360
NEET 2023 BORA VARUN CHAKRAVARTHI 720
For More Join: @IITJEE_Advanced

26. Consider the electric field E versus position x graph. What is the minimum velocity (in m/s) that should be given to a point charge $-Q$ of mass M at $x=3L$ so that it can reach the origin? Take $Q=1C$, mass of the charge M as $1kg$, $E_0=5N/C$ and $L=5m$.



27. Two coherent point sources of frequency ($f=\frac{10v}{d}$ where v is speed of light) are placed at a distance d apart as shown in figure. The receiver is free to move along the dotted line shown in the figure. Find total number of maxima observed by receiver.

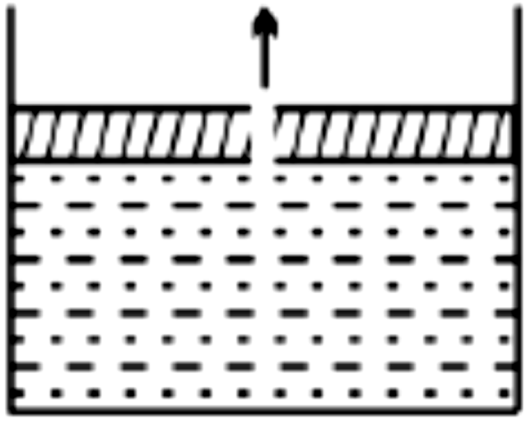


Rankers Academy JEE

28. Assuming a particle to have the form of a sphere and absorbs all incident light, find the radius $R(in\ \mu m)$ of a particle for which its gravitational attraction to the Sun is counterbalanced by the force that light exerts on it. The power of light radiated by the Sun equals $P=4\times10^{26}W$ and the density of the particle is $\rho=1.0g/cm^3$. Use $G=\frac{20}{3}\times10^{-11}Nm^2/kg^2$, $\pi=\frac{25}{8}$ and mass of the Sun = $2\times10^{30}kg$. Fill the value of $10R$ in OMR sheet.

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300/300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341/360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

29. In the figure shown, water is filled in the cylindrical tank of cross–sectional area $100 cm^2$ up to a height of 10 cm. The mass of the thin, tight fitting and frictionless piston is 100 kg. The small hole in the piston has area $2\sqrt{2}\, mm^2$. Due to weight of the piston, water spurts upwards in a jet from the hole. Find the time (in s) in which the tank becomes empty. If your answer is n, fill the value of $n/5$. (Assume that spurted water does not fall back onto the piston and density of water = $10^3 kg/m^3$)



30. A vertical rod of mass 4 kg is hanging from a rope and a 3 kg cat is grasping the bottom end of the rod. The rope is cut off and the frightened cat begins to run up on the rod. While the rod is falling vertically, the cat remains at the same height, with respect to the ground. The acceleration of the rod is $(ng/4)\, m/s^2$ where g is the acceleration due to gravity. Find n.

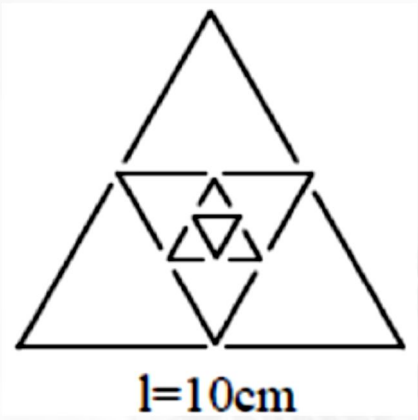
SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class 300 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class 341 360
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class 720
For More Join: @IITJEE_Advanced

An equilateral triangular fractal is made out of infinitely thin wire with linear mass density $\lambda = 100\, gm^{-1}$. An equilateral triangle with side length $\ell = 10\, cm$ is made and then add mid segments of the inner medial triangle created in the previous step and so on. The result is shown in fig.



The mass of such a fractal is M and its moment of inertia about one of the medians is I.

31. The value of M is ____ grams.

32. The value of I is ____ $\times 10^{-5} kgm^2$.

PARAGRAPH-II

One side of a thin metal plate is illuminated by the Sun. When the air temperature is T_0, the temperature of the illuminated side is T_1 and that of the opposite side is T_2. The temperatures of those sides be T_1' and T_2', if another plate of double thickness is used in place of first plate. (Take $T_0 = 300\, K, T_1 = 330\, K, T_2 = 320\, K$)

33. The value of $T_1'(K)$ is ____

34. The value of $T_2'(K)$ ____

Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. Match the appropriate shape around Xenon the right with each of the species on the left.

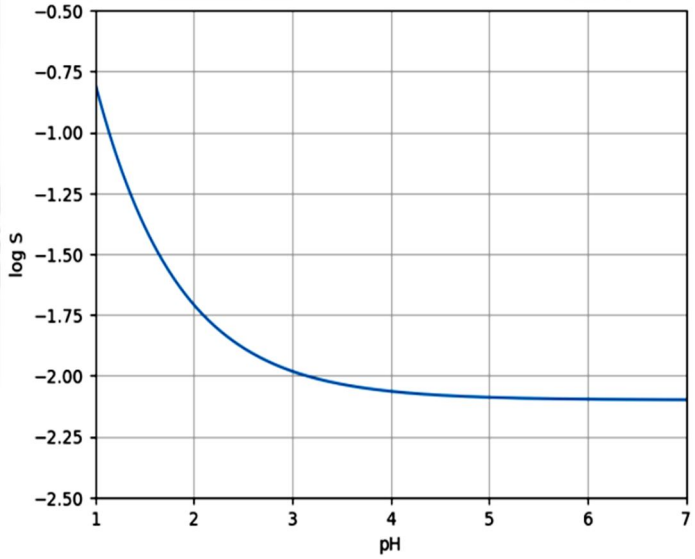
P) $FXeO(OSO_2F)$	I) Linear
Q) $FXeN(SO_2F)_2$	II) Pyramidal
R) XeO_3	III) T–shaped
S) $XeOF_2$	IV) Bent

A) P–I, Q–I, R–II, S–III B)P–I, Q–I, R–II, S–IV
C)P–IV, Q–I, R–II, S–III D) P–I, Q–IV, R–II, S–III

36. The magnitudes of CFSE in $\left[M(H_2O)_6\right]^{+n}$ for Mn and Fe ions satisfy the relations

A) $Mn^{+2} < Mn^{+3}$ and $Fe^{+2} < Fe^{+3}$ B) $Mn^{+2} > Mn^{+3}$ and $Fe^{+2} > Fe^{+3}$
C) $Mn^{+2} < Mn^{+3}$ and $Fe^{+2} > Fe^{+3}$ D) $Mn^{+2} > Mn^{+3}$ and $Fe^{+2} < Fe^{+3}$

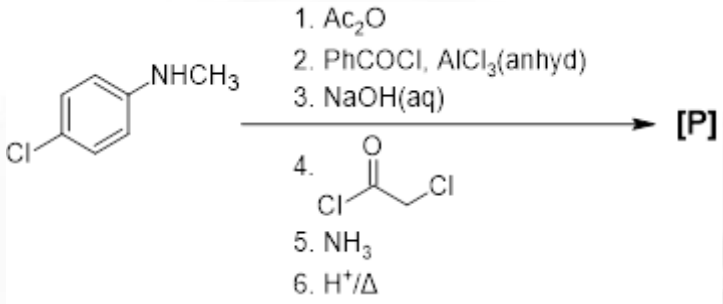
37. The graph shows the logarithm of the molar solubility S of an ionic solid as a function of pH. Which solid is it?

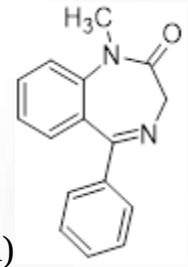


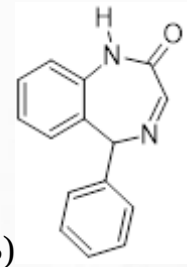
Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300/300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341/360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

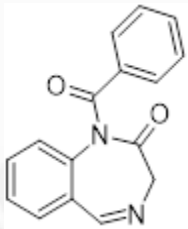
A) $AgCl$ B) PbI_2 C) BaF_2 D) $Ca(NO_3)_2$

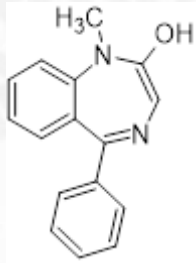
38. The major product **[P]** of the following reaction is



A) 

B) 

C) 

D) 

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. BOD (Biochemical oxygen demand) is a measure of

A) Oxygen present in water B) Oxygen required by microorganisms

C) Total dissolved solids D) P^H of water

40. The **CORRECT** statements regarding $Co_2(CO)_8$ are

A) It reacts with Na to give $Na[Co(CO)_4]$

B) It contains three bridging carbonyls

C) One metal–metal bond is present

D) No metal–metal bond is present

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class 300 300 RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class 341 360 RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class 720 RANK 1
For More Join: @IITJEE_Advanced

41. Choose the correct statement(s) regarding the role and behaviour of a catalyst in a chemical reaction:

A) A catalyst is typically not written in the overall balanced chemical equation because, although it may participate in intermediate steps, it is regenerated at the end of the reaction and therefore does not undergo any permanent chemical change.

B) The presence of a catalyst can alter the mechanism of the reaction and, as a consequence, may change the form of the rate law that experimentally describes the reaction kinetics.

C) In certain situations, the addition of a catalyst can lead to a reduction in the rate of a particular reaction pathway.

D) A catalyst can modify an existing reaction pathway to make it faster or provide an alternative pathway altogether, both of which involve a lower activation energy compared with the uncatalyzed process.

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

42. The sum of the number of shared corners of the constituent SiO_4 units in orthosilicate, pyrosilicate, cyclicsilicate and sheetsilicates is

43. Observe the following reactions

$$Sn + PC\ell_5 \longrightarrow A + B \quad (1)$$

$$AgNO_3 + H_2O + H_3PO_2 \longrightarrow Ag + D + E \quad (2)$$

(Reactions are not balanced)

B and E are phosphorous containing products.

x is oxidation state of tin in A.

y is oxidation state of phosphorous in E.

What is x + y?

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026


For More Join: @IITJEE_Advanced

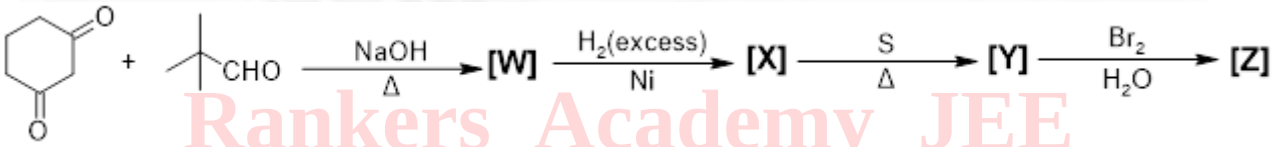
44. 20.4 g of KHP $\left(KC_8H_5O_4\right)$ is dissolved in water, giving a solution with a mass of 118.4 g. A solution of sodium hydroxide is also prepared. 4.4 g of this sodium hydroxide solution reacts completely with 5.8 of the KHP solution above. Sodium hydroxide reacts in a 1:1 mole ratio with KHP. In a similar reaction, 4.4 g of the sodium hydroxide solution reacts completely with a 5.9 g sample of vinegar (containing acetic acid, CH_3COOH). Sodium hydroxide reacts in a 1:1 mole ratio with acetic acid.Calculate the percentage by mass of acetic acid present in the vinegar sample.
[Molar mass of KHP = 204, $NaOH$ = 40 and CH_3COOH = 60]

45. The rate law for the reaction of OH^- with *tert*-butyl bromide to form an elimination product in 75% ethanol 25% water at 30 °C is the sum of the rate laws for the E2 and E1 reactions:

rate = 7.1×10^5 [*tert*-butyl bromide] $\left[OH^-\right]+1.5\times10^{-5}$ [*tert*-butyl bromide]

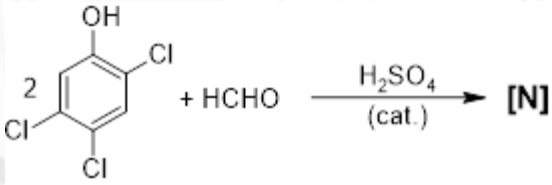
For what concentration (mol/L) of OH^- does the reaction proceed through 96 % E2 mechanism?

46. Consider the following reaction sequence:



The sum of the number of sp^3 carbon atoms and the number of heteroatoms in **[Z]** is

47. The molar mass of the final product [N] of the following reaction is



[Given: Atomic mass of H = 1, C = 12, O = 16, Cl = 35.5]

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

The rate of the gas-phase reaction.

$H_2 + I_2 \longrightarrow 2HI$ is given by $rate = -\frac{d[I_2]}{dt} = k[H_2][I_2]$

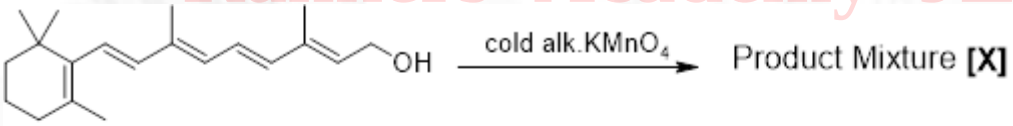
with k = 0.2 L $mol^{-1}s^{-1}$ at $400^\circ C$. The reaction is set up by taking 0.8 mol of H_2 and 0.4 mol of I_2 in a 1 L flask at $400^\circ C$. Let the initial rate at which heat is absorbed or emitted during the reaction be $x\ kJ\ s^{-1}$ and the total heat that would have been evolved at infinite time be $y\ kJ$. Assume that the enthalpy change at $400^\circ C$ is the same as that at $25^\circ C$. [Given bond dissociation energies (in KJ mol^{-1} at $25^\circ C$): H – H = 435; I – I = 150 & H – I = 300]

48. What is the value of x?

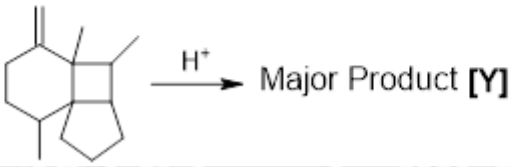
49. What is the value of y?

PARAGRAPH-II

Consider the following reactions:



Vitamin A



50. How many chiral compounds are present in product mixture **[X]**?

51. What is the sum of the number of methyl groups and number of 5 membered rings in the major product **[Y]**?

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300/300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341/360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

41 Years of Excellence 1986-2026

LEADING BY MILES SRI CHAITANYA DOMINATES

JEE ADVANCED 2025

29 Ranks in Top 100 in All-India Open Category



Ramaiah Academy JEE

4 Students in Top 11 in JEE-Advanced 2025, All India Open Category

16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS: 29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS: 113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS: 205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT: 745	NUMBER OF QUALIFIED RANKS: 4,212

040 66 15 15 15 www.srichaitanya.net

For More Join: @IITJEE_Advanced